

# Learning mixtures of teachers: main question

We study a high-dimensional data model where each sample is generated by one of two hidden linear teachers.

The input  $X$  contains partial information about the hidden teacher: when the two data components are more separated, the teacher identity becomes easier to infer.

## Main question:

How do neural networks use this information during learning?

We compare two architectures:

two-layer MLP

and

Mixture-of-Experts.

Using statistical-physics methods, we analyze:

**Population dynamics:** stable learning branches and initialization dependence,

**Finite samples:** ERM, generalization, and recovery thresholds.

## Data model

$$k_+^*, k_-^* \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}\left(0, \frac{1}{D} I_D\right), \quad c^* \sim \mathcal{N}\left(0, \frac{\gamma}{D} I_D\right).$$

For each sample,

$$\varepsilon \in \{\pm 1\}, \quad P(\varepsilon = +1) = P(\varepsilon = -1) = \frac{1}{2},$$

$$\tilde{X} \sim \mathcal{N}(0, I_D), \quad X = \tilde{X} + \varepsilon c^*,$$

and

$$y = \langle \tilde{X}, k_\varepsilon^* \rangle.$$

Introduce the three planted directions

$$k_{\parallel} = \frac{k_+^* + k_-^*}{2}, \quad k_{\perp} = \frac{k_+^* - k_-^*}{2}, \quad u_c = \frac{c^*}{\sqrt{\gamma}}.$$

For large  $D$ ,  $k_{\parallel} \perp k_{\perp} \perp u_c$ ,  $\|k_{\parallel}\|^2 \simeq \|k_{\perp}\|^2 \simeq \frac{1}{2}$ ,  $\|u_c\|^2 \simeq 1$ .

### Interpretation

$k_{\parallel}$  is the common teacher component,  $k_{\perp}$  distinguishes the two teachers, and  $u_c$  contains information about which teacher generated the sample.

# Two-expert Mixture of Experts

The student is

$$\hat{y}(X) = s_1(X)\langle k_1, X \rangle + s_2(X)\langle k_2, X \rangle,$$

with

$$(s_1, s_2) = \text{softmax}(\langle g_1, X \rangle, \langle g_2, X \rangle).$$

Introduce average and difference coordinates

$$\bar{k} = \frac{k_1 + k_2}{2}, \quad d = \frac{k_1 - k_2}{2}, \quad g_\Delta = g_1 - g_2.$$

Then the prediction becomes

$$\hat{y}(X) = \langle \bar{k}, X \rangle + \tanh\left(\frac{\langle g_\Delta, X \rangle}{2}\right) \langle d, X \rangle.$$

## Interpretation

$\bar{k}$  learns the common teacher component,  $d$  describes how the experts differ, and  $g_\Delta$  determines the routing.

# Population dynamics: two low-dimensional branches

Although

$$\bar{k}, d, g_{\Delta} \in \mathbb{R}^D,$$

the balanced population dynamics organizes into simple low-dimensional geometries. For  $p = 1/2$  and  $D \rightarrow \infty$ , I find two main invariant branches.

## Routing branch

$$\bar{k} = Ak_{\parallel}, \quad d = Bk_{\perp}, \quad g_{\Delta} = Cu_c.$$

Here the gate learns the cluster direction  $u_c$ , while the expert difference learns  $k_{\perp}$ .

## Factor-swapped branch

$$\bar{k} = Ak_{\parallel}, \quad d = Bu_c, \quad g_{\Delta} = Ck_{\perp}.$$

Here the roles of  $u_c$  and  $k_{\perp}$  are exchanged.

## Main reduction

The high-dimensional vector dynamics reduces to a few scalar amplitudes.

## Routing branch: generic trajectories reach a larger manifold first

The pure routing geometry is

$$\bar{k} = Ak_{\parallel}, \quad d = Bk_{\perp}, \quad g_{\Delta} = Cu_c.$$

However, generic full- $D$  trajectories do not reach this manifold immediately.

The expert directions align first, while the gate keeps an additional component outside the planted three-dimensional signal space:

$$\bar{k} = Ak_{\parallel}, \quad d = Bk_{\perp}, \quad g_{\Delta} = Cu_c + r_g,$$

with

$$r_g \perp \text{span}\{k_{\parallel}, k_{\perp}, u_c\}.$$

Define only its magnitude

$$R = \|r_g\|.$$

Therefore the extended routing dynamics is described by

$$A(t), \quad B(t), \quad C(t), \quad R(t).$$

$$\text{generic trajectory} \longrightarrow \text{extended routing} \longrightarrow \text{pure routing } (R = 0).$$

## Extended routing: four-dimensional dynamics

Substituting

$$\bar{k} = Ak_{\parallel}, \quad d = Bk_{\perp}, \quad g_{\Delta} = Cu_c + r_g$$

into the population gradient flow gives a closed dynamical system for

$$\boxed{A(t), \quad B(t), \quad C(t), \quad R(t).}$$

The first three coordinates describe the planted routing structure.

The new coordinate

$$R = \|r_g\|$$

measures how much of the gate is still outside the signal space.

Its evolution has the multiplicative form

$$\boxed{\dot{R} = -\lambda_R(A, B, C, R; \gamma) R.}$$

Therefore, on the routing trajectories  $R(t) \rightarrow 0, (t \rightarrow \infty)$

### Main point

The old three-dimensional routing manifold is the asymptotic limit of a larger four-dimensional manifold that describes the long transient seen in the full model.

## The final routing relaxation becomes exponentially slow

$$R(t) = R(t_0) \exp \left[ - \int_{t_0}^t \lambda_R(s) ds \right].$$

Thus the convergence of the gate to the pure routing manifold is controlled entirely by the transverse relaxation rate  $\lambda_R$ .

In the final routing regime, for large  $\gamma$ ,

$$\lambda_R \sim \lambda_{\perp}(\gamma) \propto \frac{e^{-\gamma/2}}{\sqrt{\gamma}}.$$

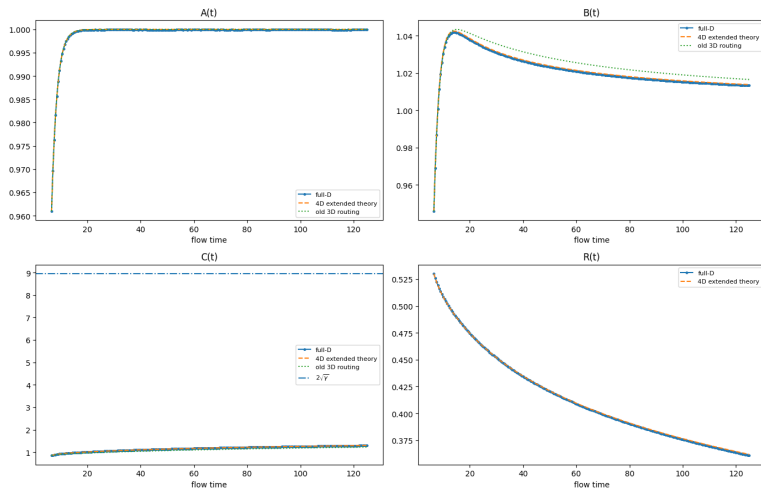
Therefore

$$t_{\text{relax}} \propto \sqrt{\gamma} e^{\gamma/2}.$$

### Interpretation

The experts and the signal part of the gate can organize much earlier, while the remaining off-signal gate component decays on an exponentially longer time scale.

# Extended routing: reduced dynamics vs full model



The full  $D$ -dimensional dynamics is represented by only

$$A(t), B(t), C(t), R(t).$$

## Observation

The four-dimensional theory follows the full- $D$  trajectory, including the nonzero transverse gate amplitude  $R(t)$ .

The old routing description corresponds to imposing  $R = 0$  and therefore misses the long transient.



# Extended routing in the original order parameters

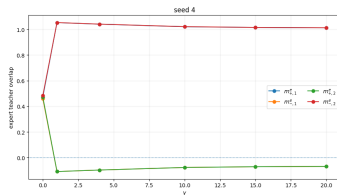
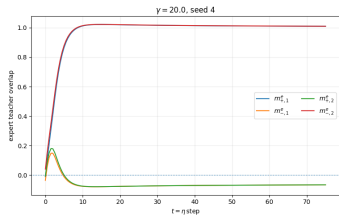
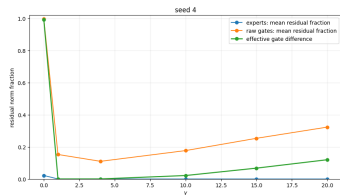
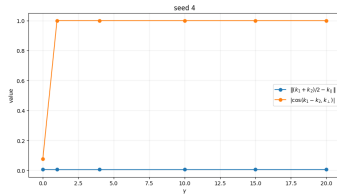
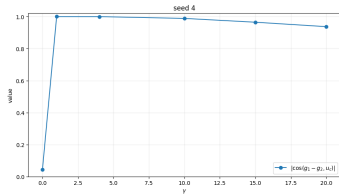
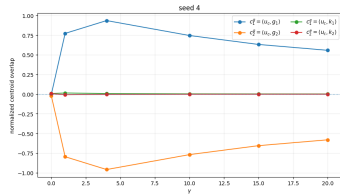


Figure:  $m_{ti}^e := \langle k_t^*, k_i \rangle$ ,  $m_{ti}^g := \langle k_t^*, g_i \rangle$  and  $c_i^e := \langle u_c, k_i \rangle$ ,  $c_i^g := \langle u_c, g_i \rangle$

## Factor-swapped population branch

The second exact population geometry is

$$\bar{k} = Ak_{\parallel}, \quad d = Bu_c, \quad g_{\Delta} = Ck_{\perp}.$$

Compare it with routing:

|                | $d$         | $g_{\Delta}$ |
|----------------|-------------|--------------|
| routing        | $k_{\perp}$ | $u_c$        |
| factor-swapped | $u_c$       | $k_{\perp}$  |

### Interpretation

The same two informative directions are learned, but they are assigned to different parts of the MoE: routing puts cluster information in the gate, whereas the swapped solution puts teacher-difference information in the gate.

## Factor-swapped branch: asymptotic solution

The factor-swapped dynamics also closes on three amplitudes

$$A(t), \quad B(t), \quad C(t).$$

The common component converges to

$$A \rightarrow 1.$$

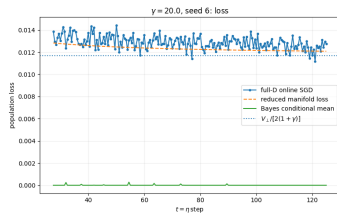
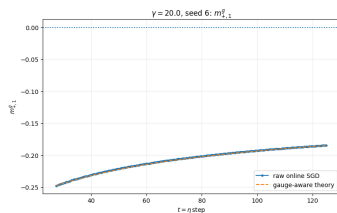
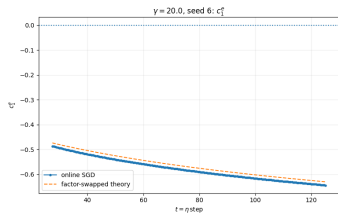
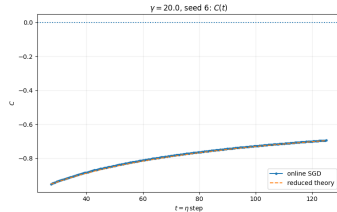
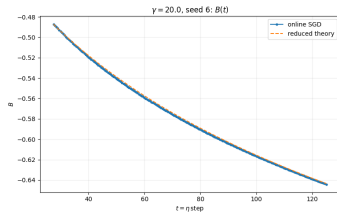
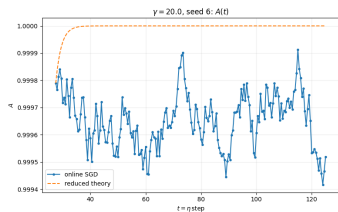
Inside this manifold, the optimal solution is scale-separated:

$$C \rightarrow 0, \quad B \rightarrow \infty, \quad BC \rightarrow \frac{2\sqrt{\gamma}}{1+\gamma}.$$

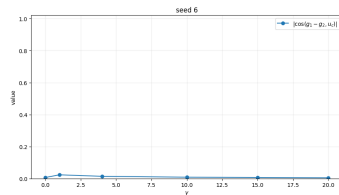
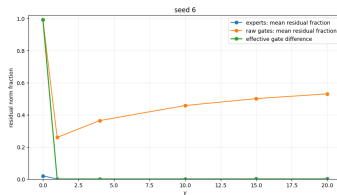
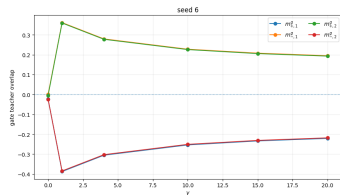
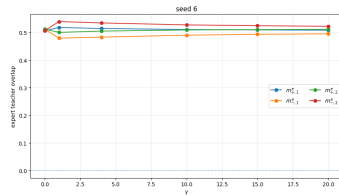
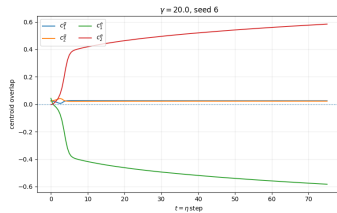
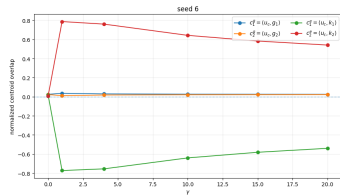
The corresponding population loss approaches

$$\inf \mathcal{E}_{\text{swap}} = \frac{1}{4(1+\gamma)}.$$

# Factor-swapped in the original order parameters



# Factor-swapped in the original order parameters



# What in the initialization selects the MoE branch?

At initialization, decompose only the two vectors responsible for the branch competition:

$$d = B_{\parallel} k_{\parallel} + B_k k_{\perp} + B_c u_c + r_d,$$

$$g_{\Delta} = C_{\parallel} k_{\parallel} + C_k k_{\perp} + C_c u_c + r_g.$$

The four branch-selecting projections are

$$B_k = \frac{\langle d, k_{\perp} \rangle}{\|k_{\perp}\|^2}, \quad B_c = \langle d, u_c \rangle, \quad C_k = \frac{\langle g_{\Delta}, k_{\perp} \rangle}{\|k_{\perp}\|^2}, \quad C_c = \langle g_{\Delta}, u_c \rangle.$$

For isotropic initialization  $k_i^{(0)} \sim \sigma_e \mathcal{N}\left(0, \frac{I_D}{D}\right)$ ,  $g_i^{(0)} \sim \sigma_g \mathcal{N}\left(0, \frac{I_D}{D}\right)$ , with  $\sigma_e = \sigma_g = \sigma$ ,

$$\|r_d^{(0)}\|^2 \xrightarrow{D \rightarrow \infty} \frac{\sigma^2}{2}, \quad \|r_g^{(0)}\|^2 \xrightarrow{D \rightarrow \infty} 2\sigma^2, \quad \xi := \langle r_g^{(0)}, X \rangle \sim \mathcal{N}(0, 2\sigma^2).$$

At linear order  $(B_{\parallel}, C_{\parallel})$  decouple from branch selection

## Two equivalent unstable branch modes

The four branch-selecting projections split into two closed sectors:

$$(B_k, C_c) \leftrightarrow \text{extended routing}, \quad (B_c, C_k) \leftrightarrow \text{factor-swapped}.$$

Define

$$x_{\text{ext}} = \begin{pmatrix} B_k/\sqrt{2} \\ C_c/2 \end{pmatrix}, \quad x_{\text{swap}} = \begin{pmatrix} B_c \\ C_k/(2\sqrt{2}) \end{pmatrix}.$$

For small equal initialization their initial dynamics is

$$\dot{x}_{\text{ext}} = J_{\text{ext}}x_{\text{ext}}, \quad \dot{x}_{\text{swap}} = J_{\text{swap}}x_{\text{swap}},$$

with

$$J_{\text{ext}} = \begin{pmatrix} -\frac{a}{2} & \kappa \\ \kappa & (1+\gamma)m \end{pmatrix}, \quad J_{\text{swap}} = \begin{pmatrix} -\frac{(1+\gamma)a}{2} & \kappa \\ \kappa & m \end{pmatrix}.$$

Here  $a = \mathbb{E}\left[\tanh^2\left(\frac{\xi}{2}\right)\right]$ ,  $\kappa = \frac{1-a}{2}\sqrt{\frac{\gamma}{2}}$ ,  $m = -\frac{a}{2} + O(\sigma^4)$ .

$$J_{\text{swap}} = PJ_{\text{ext}}P + O(\sigma^4), \quad P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

so the two sectors have the same leading eigenvalue  $\lambda_+ = -\frac{(\gamma+2)a}{4} + \sqrt{\kappa^2 + \frac{\gamma^2 a^2}{16}}$ .

### Consequence

Neither branch grows faster initially: branch selection comes from the initial projection onto the two unstable modes.

# Initialization predicts the selected branch

The unstable eigenvectors are

$$v_{\text{ext}}^+ \propto \begin{pmatrix} 1 \\ \rho \end{pmatrix}, \quad v_{\text{swap}}^+ \propto \begin{pmatrix} \rho \\ 1 \end{pmatrix},$$

with  $\rho = \frac{\sqrt{\kappa^2 + \gamma^2 a^2 / 16} - \gamma a / 4}{\kappa}$

Hence the unstable amplitudes already present at initialization are

$$U_{\text{ext}}^{(0)} = \frac{B_k^{(0)}}{\sqrt{2}} + \rho \frac{C_c^{(0)}}{2}, \quad U_{\text{swap}}^{(0)} = \rho B_c^{(0)} + \frac{C_k^{(0)}}{2\sqrt{2}}.$$

Both grow with the same exponent:

$$U_{\text{ext}}(t) \simeq U_{\text{ext}}^{(0)} e^{\lambda_+ t}, \quad U_{\text{swap}}(t) \simeq U_{\text{swap}}^{(0)} e^{\lambda_+ t}.$$

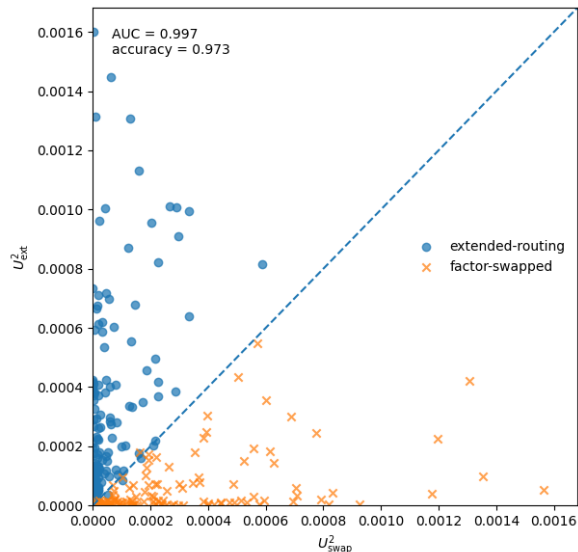
Therefore

$$\Delta_{\text{init}} = \left( \frac{B_k^{(0)}}{\sqrt{2}} + \rho \frac{C_c^{(0)}}{2} \right)^2 - \left( \rho B_c^{(0)} + \frac{C_k^{(0)}}{2\sqrt{2}} \right)^2$$

$$\Delta_{\text{init}} > 0 \Rightarrow \text{extended routing}, \quad \Delta_{\text{init}} < 0 \Rightarrow \text{factor-swapped}.$$



# Initialization-dependent selection between the two MoE branches.



For random isotropic initializations at  $\gamma = 5$ , each point shows

$$U_{\text{swap}}^2 \quad \text{vs} \quad U_{\text{ext}}^2.$$

The diagonal

$$U_{\text{ext}}^2 = U_{\text{swap}}^2$$

is the leading-order theoretical decision boundary.

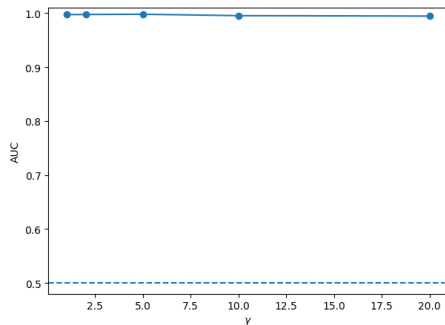
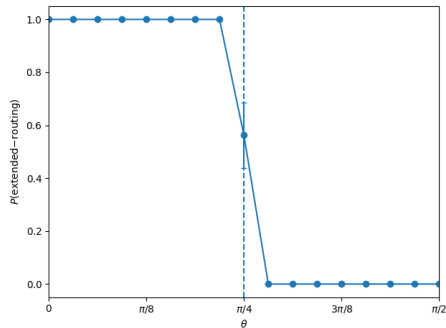
Above the diagonal:

$$U_{\text{ext}}^2 > U_{\text{swap}}^2 \Rightarrow \text{extended routing.}$$

Below the diagonal:

$$U_{\text{swap}}^2 > U_{\text{ext}}^2 \Rightarrow \text{factor-swapped.}$$

# Population dynamics: branch selection



**Figure:** **(a)** Controlled competition between the two unstable modes. Initializations are constructed with fixed total unstable power,  $U_{\text{ext}}^2 + U_{\text{swap}}^2 = R^2$ , and parameterized as  $U_{\text{ext}} = R \cos \theta$ ,  $U_{\text{swap}} = R \sin \theta$ . The empirical probability of converging to the extended-routing branch changes around the predicted boundary  $\theta = \pi/4$ , where the two unstable amplitudes are equal. **(b)** Predictive performance of the leading-order initialization theory across routing strengths. We report the ROC AUC of the score  $U_{\text{ext}}^2 - U_{\text{swap}}^2$  as a function of  $\gamma$

# Finite-sample ERM

Population dynamics assumes access to the data distribution.  
Now use a fixed dataset with

$$N = \alpha D$$

samples and minimize

$$\mathcal{L}_{\text{ERM}} = \frac{1}{2N} \sum_{\mu=1}^N (y^\mu - \widehat{y}(X^\mu))^2 + \frac{\lambda}{2} (\|\bar{k}\|^2 + \|d\|^2 + \|g_\Delta\|^2).$$

## Finite-sample order parameters

$$W = \begin{pmatrix} \bar{k}^\top \\ d^\top \\ g_\Delta^\top \end{pmatrix}, \quad V = \begin{pmatrix} k_\parallel^\top \\ k_\perp^\top \\ u_c^\top \end{pmatrix}.$$

$$\boxed{M = WV^\top, \quad q = WW^\top, \quad G = VV^\top.}$$

The part that cannot be explained by the planted subspace is

$$\boxed{R_{\text{tr}} = q - MG^{-1}M^\top.}$$

Equivalently, if

$$W_i = W_i^\parallel + r_i, \quad r_i \perp \text{span}\{k_\parallel, k_\perp, u_c\},$$

then

$$\boxed{(R_{\text{tr}})_{ij} = \langle r_i, r_j \rangle.}$$

### Interpretation

$M$  describes the learned planted geometry, while  $R_{\text{tr}}$  measures the finite-sample components outside the teacher–centroid subspace.

# Finite- $\alpha$ continuation of the population branches

The finite-sample ERM solutions retain the same planted geometries.

## Routing

$$\bar{k} \simeq Ak_{\parallel}, \quad d \simeq Bk_{\perp}, \quad g_{\Delta} \simeq Cu_c + r_g.$$

The dominant residual is in the gate:

$$R_{33} > 0, \quad \frac{R_{33}}{\text{Tr } R_{\text{tr}}} \simeq 1.$$

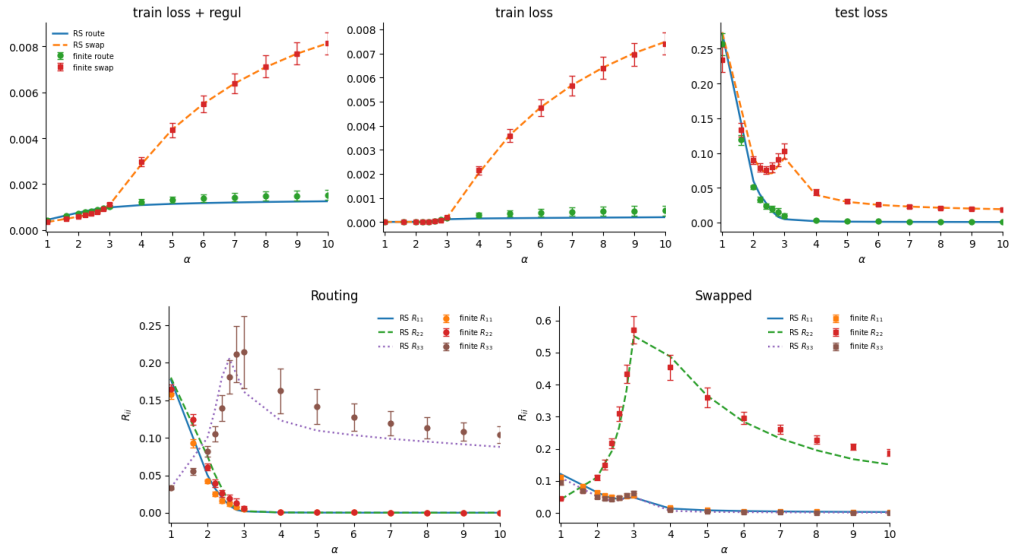
## Factor-swapped

$$\bar{k} \simeq Ak_{\parallel}, \quad d \simeq Bu_c + r_d, \quad g_{\Delta} \simeq Ck_{\perp}.$$

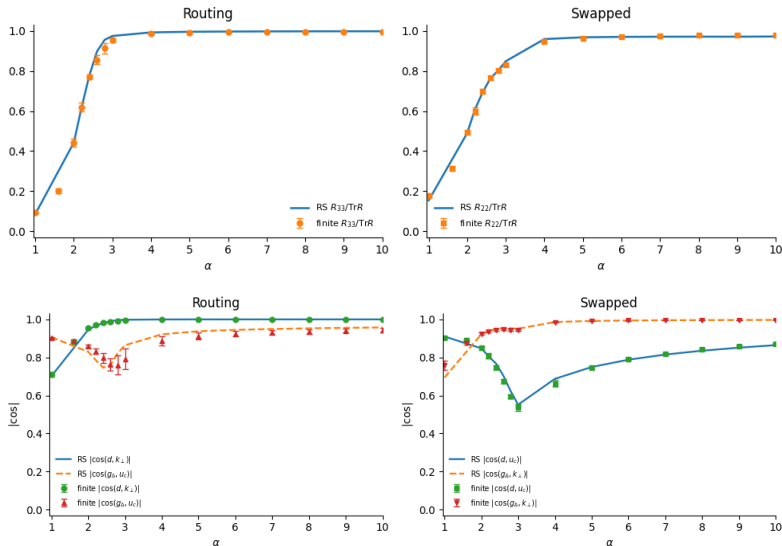
$$R_{22} > 0, \quad \frac{R_{22}}{\text{Tr } R_{\text{tr}}} \simeq 1.$$

Finite data preserve the population branch geometry, but populate a branch-specific transverse direction.

# Finite-sample ERM at $\gamma = 20$



# Finite-sample ERM at $\gamma = 20$



## Two-layer MLP student

We also study a two-layer MLP trained on the same teacher-mixture data.

$$\hat{y}(X) = \sum_{i=1}^F a_i \sigma(\langle k_i, X \rangle) + b_0.$$

Here

$k_i \in \mathbb{R}^D$  are first-layer weights,  $a_i \in \mathbb{R}$  are output weights.

For the main population experiments we take

$$F = 2, \quad \sigma(x) = \text{ReLU}(x).$$

### Goal

Describe which teacher directions are learned by the hidden neurons, and whether the network converges to an intuitive teacher-separating solution or to another stable branch.



# MLP polarized

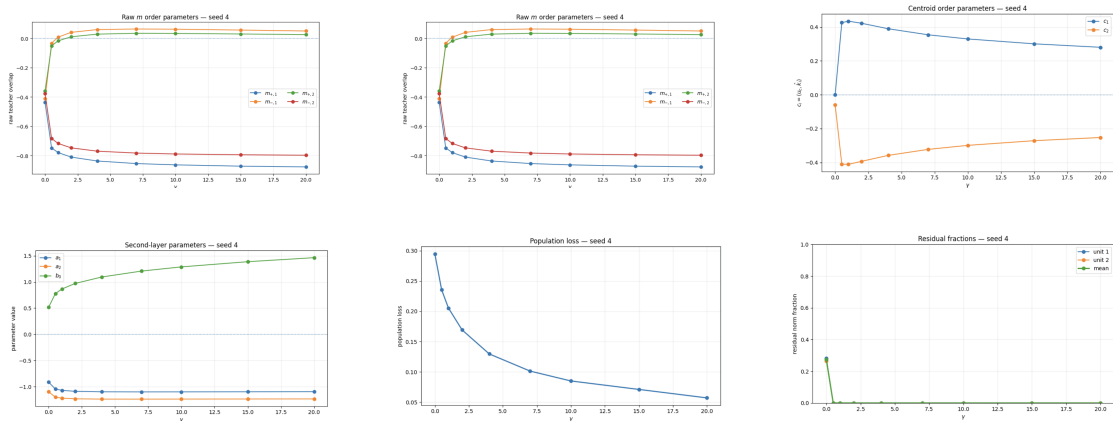
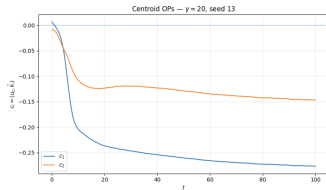
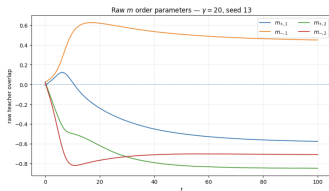
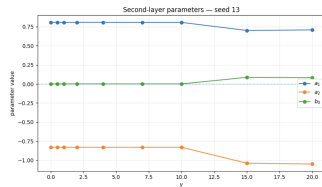
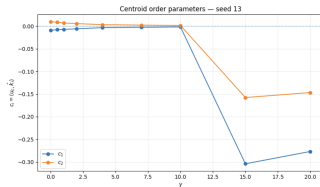
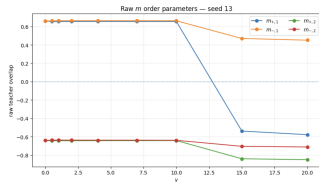
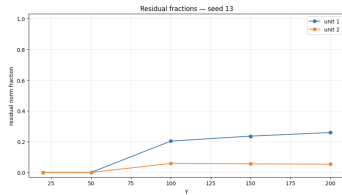
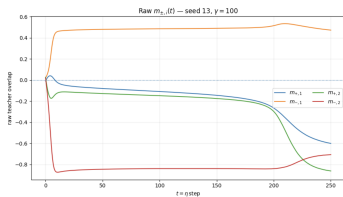
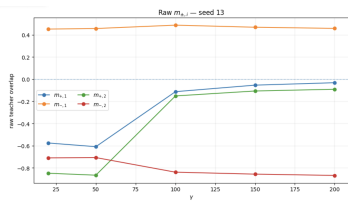


Figure: Polarized branch, dependence on  $\gamma$

# MLP another branch



For small  $\gamma$ , the system converges to a solution where the average of the teachers is learned:  $r = 0$ ,  $c$  close to zero, and the loss is around 0.25. This corresponds to the regime described earlier in the theory. As  $\gamma$  increases, a transition occurs to a different solution (though the threshold value of  $\gamma$  varies and can be quite large; on the seed 13 plot, the transition starts at  $\gamma = 15$ ). In this regime, the model no longer learns simply the average of the teachers, and  $r = 0$ .



**Figure:** Upon an even larger increase in  $\gamma$ , the system enters a regime where only one teacher is learned. After running experiments for the large values of  $\gamma$  described above with more training steps, we found that under sufficiently long training, the transition often occurs from the third regime back to the second one.

# Main takeaways

- ▶ **Mixture-of-Experts exhibits two stable population branches:** the intuitive routing solution and a non-intuitive factor-swapped solution.
- ▶ **Branch selection is initialization-dependent:** the projections of the initial weights onto two unstable modes predict which branch is reached by gradient flow.
- ▶ **Finite-sample ERM preserves the population geometry:** the two branches remain visible at finite  $\alpha$ , while finite data generate branch-specific transverse residual components.
- ▶ **MLPs also exhibit multiple stable learning regimes:** depending on data informativeness and initialization, the network can converge to qualitatively different solutions.